FLPK-BiSeNet: Federated Learning Based on Priori Knowledge and Bilateral Segmentation Network for Image Edge Extraction

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***Abstract*—Federated learning can effectively ensure data security and improve the problem of data islanding. However, the performance of federated learning-based schemes could be better due to the imbalance of image data. Therefore, this paper proposes a federated learning approach based on priori knowledge and a bilateral segmentation network for image edge extraction. First, federated learning can distribute training images for some special complex images due to the small sample and unshared data. Then, the image with similar edge information to the original image is learned to obtain prior knowledge, and the local uniform sparsity method is used to strengthen the detail features and weaken the background features. Based on the bilateral segmentation network, we introduce a dilated pyramid pooling layer and multi-scale feature fusion module to fuse the shallow detailed features in the context path with the deep abstract features obtained through the dilated pyramid pooling. The final result is obtained by fusing the result with prior knowledge and the result with the context path. Finally, we conduct experiments on some public datasets, and the results show that the proposed method greatly improves extraction accuracy compared with the traditional and the most advanced methods.**

***Index Terms*—Federated learning, priori knowledge, image edge extraction, bilateral segmentation network.**

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I. Introduction

WITH the rise of artificial intelligence, digital image processing is more widely used. Furthermore, image edge extraction directly affects the quality of image processing results, such as face recognition, license plate positioning, remote sensing image processing, underwater image recognition and so on [1], [2], [3]. The purpose of edge extraction is to separate the target information from the background information in the image, reduce the complexity of the actual project, and improve the efficiency of information processing. Therefore, the research of edge extraction is of great significance [4], [5].

At present, many scholars have proposed different methods for edge extraction algorithms [6], [7]. The commonly used edge extraction algorithms in image processing include the Sobel and Prewitt operators, using first-order partial derivatives and the Log operator, using second-order partial derivatives. However, these algorithms are simple and need help overcoming incomplete, discontinuous and jitter edges. The Canny operator is more complete than the previous operators in edge extraction, but under the interference of strong Gaussian noise, the edge information cannot be completely extracted [8]. Therefore, Luo et al. [9] proposed an edge extraction method based on graph theory, which regarded the image as an undirected graph and determined the mean value of ownership value as a threshold. This method could improve the image jitter problem but could not extract the ideal edge information for the image with strong noise. Kumawat and Panda [10] proposed the improved Canny operator fused with fuzzy enhancement and extracted image edges by constructing the segmented fuzzy membership function to distinguish between high and low gray levels. Although it could effectively achieve the purpose of edge refinement, it also had no good extraction effect for images polluted by high noise.

In addition to the detection operator, the fuzzy set theory can also be used for image edge extraction [11]. However, this method requires much computation time while bringing a good edge extraction effect, which is difficult to apply to high real-time scenes. For example, Edge extraction is closely related to image features, most of which are based on the probability distribution of joint images [12]. Local features such as texture, illumination and brightness are input into the network classifier for edge determination to improve the ability of edge extraction [13]. However, although the network performance of this kind of method has been improved, the experimental process is wasteful and costly. Furthermore, technological processes are more complex, with poor real-time performance.

In recent years, deep learning with Convolutional Neural Networks (CNN) as the core technology has been widely used in many fields [14], [15], which emphasizes the importance of automatic hierarchical feature learning and effectively improves the performance of edge detection. Compared with the traditional computational learning theory, which merely uses the features of the final output layer, the multi-scale fusion network adopts the feature learning method of multi-scale and multi-level, which significantly improves the edge detection effect based on the Visual Geometry Group (VGG16) network [16]. The edge detection performance is improved using richer convolution features and more robust loss functions. However, the CNN module is still used in the multi-scale fusion network, and the sensing sampling area is a fixed geometric structure, which leads to the limited feature extraction ability of geometric transformation and modeling.

In the image edge extraction task, the DeepUNet network architecture was designed in [17]. The upper and lower sampling modules were designed to replace the convolution layer of the original U-NET encoding and decoding structure, promoting RSIs' segmentation precision. Reference [18] used the powerful feature extraction ability of dense, deep separable networks to construct a segmentation network architecture that could extract high-dimensional features of images and achieve relatively accurate segmentation of image target regions. Different from traditional detection algorithms, the detection algorithms based on deep learning do not require a complex modeling process for image object edge detection tasks. Therefore, it has aroused research interest. Based on the SSD detection framework, the authors in [19] proposed using an attention mechanism to optimize image object features in complex scenes to improve network detection performance. According to the authors in [20], hierarchical and progressive detection was carried out in combination with scene classification, which improved the accuracy and speed of target detection. In [21], a deep neural network based on an adaptive adjustment machine was proposed to improve ship detection performance by combining rotation and single-stage detection. In [22], the authors proposed a feature balance and optimization network based on anchor-free frame setting and designed to use attention to guide feature extraction at different levels of the balancing pyramid. In [23], the authors designed top-down and bottom-up combined feature fusion methods to improve the performance of multi-scale and small-scale target detection in complex backgrounds. However, the above methods ignored the global features. At present, image object edge detection methods with deep learning methods are mostly based on the Anchor mechanism, which predefines candidate boxes of different scales and aspect ratios in each pixel of the feature map [24]. However, when these methods solve the target detection problem in large complex scenes, the setting of dense candidate boxes will consume many computing resources, seriously affecting detection efficiency and performance. In contrast, the network architecture of the anchor-free method is more concise, and the anchor-free design idea will greatly reduce the occupied computing resources, which is more suitable for large scene target detection in practical applications.

Traditional detection methods require manual feature design and complicated parameter adjustment. This algorithm needs better adaptability under complex scenes. The existing detection methods based on deep learning can improve detection automation and intelligence, but some things could still be improved in large complex scenes.

1) First, the image size of the large scene is large, while the target presents an obvious sparse distribution phenomenon, and the relative size is small, so the whole scene image cannot be directly sent to the network for processing.
2) Second, in complex scenes, land targets have great influence, and direct block processing and sending to the network for detection will lead to more land false alarms and decreased detection performance.
3) Third, the target occupies a relatively small proportion in the large scene. The algorithm's efficiency will be reduced if the large scene image is traversed directly to search for the target.

We present a federated learning approach based on priori knowledge and a bilateral segmentation network for edge detection to solve the above problems. Our main contributions are as follows:

1) Federated learning architecture is used to break the barrier of "data island"; it carries out distributed training and sharing of data, which can realize image features rich management.
2) The texture features are recorded by learning the same kind of high-quality images. The local uniform sparsity method is used to strengthen the target features and weaken the background features, which overcomes the threshold dependence of the edge extraction algorithm.
3) Based on the BiSeNet network structure, a dilated pyramid model is introduced into the light feature extraction network in the context path to obtain deep abstract features.
4) A multi-scale feature fusion module is used to fuse the shallow detail features and deep abstract features of the light feature extraction network to obtain enhanced context information features. Through the improved context path, better edge extraction results can be obtained.

In this paper, we describe the related works about edge extraction methods in Section II. Section III detailedly explains the proposed FLPK-BiSeNet. In Section IV, we demonstrate the novel algorithm by using some natural images. A conclusion is summarized in Section V.

II. Related Works

Edge extraction technology is an image processing and analysis technology in image region segmentation, target region recognition and region shape extraction. It is an indispensable part of the machine vision system [25]. The region and edge of the image are extracted from the natural image to extract the range of the object and the prominent area information in visual perception, to hold the features and key area information satisfactorily.

Deep learning is widely used in image processing due to its powerful ability of image feature extraction. For example, FCN (Fully Convolutional Networks) method is used for image semantic segmentation, which extends the image-level classification to pixel-level classification, laying the foundation for semantic segmentation. Current edge extraction methods are mainly divided into two categories. One is to achieve the fine pixel-level classification of images by integrating high-level semantic and low-level spatial information through an encoder-decoder structure. The other one is to use the dilated convolution structure, omit the pooling layer, and reduce the loss of the input image location information in the feature extraction period. Representative methods include U-Net, SegNet, etc. The representative methods include DeepLabv3+, PSPNet, etc. Scholars have recently applied deep learning methods to image segmentation/edge extraction tasks. In [26], the authors proposed DeepUNet based on the U-NET network, designed the DownBlock module and UpBlock module to replace the convolutional layer in the encoding-decoding structure and obtained more accurate segmentation results in the optical remote sensing image segmentation task. Reference [27] proposed a segmentation network architecture based on dense deep separated convolution, which extracted image high-dimensional features through dense separated convolution and expanded convolution, and constructed an up-sampling decoding module based on bilinear interpolation to realize remote sensing image segmentation.

The existing image segmentation or edge detection algorithms based on deep learning mostly adopt a U-shaped network structure, which gradually restores the resolution of the feature map to the original image size by fusing the features of different levels of the backbone network layer by layer. However, this method will introduce too much computation in the high-dimensional feature map extraction process, so the segmentation speed is slow. The Bilateral Segmentation Network (BiSeNet) proposed in [28] can effectively balance the segmentation speed and accuracy and perform well in the semantic Segmentation of natural scene images. However, for the remote sensing image segmentation task, the feature extraction path of the bilateral network is difficult to effectively extract the image's contextual semantic and spatial information, so the segmentation effect is poor. Therefore, this paper improves BiSeNet.

A. Federated Learning

In 2016, Google proposed a new artificial intelligence technology called Federated Learning (FL) [29] to solve the problem of Android mobile phone terminal users updating

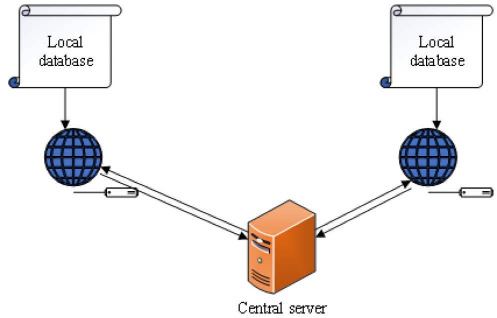


Fig. 1. Traditional Federated Learning.

models locally. FL enables multi-party organizations to realize AI cooperation without sharing data and ensures the privacy and security of user data. As shown in Fig. 1, several central servers handle traditional FL with collaboration. The local device only adopts locally owned data for training neural networks in a central server. Then the updated parameters will be obtained and sent to the coordinator.

Currently, federation learning is divided into horizontal, vertical, and federation transfer learning according to the difference between data feature space and sample space [30]. Horizontal federated learning mainly applies when the data share the same sample space, but the feature space differs. Vertical federated learning is opposite to horizontal federated learning. Federated transfer learning is used when the feature and sample spaces are different. At present, Federated Learning has started to explore various fields, such as the risk control model jointly modeled by several banking institutions in the financial field; Effectively improving the matching efficiency of information and resources in the field of smart retail; As the data features are the same and the sample data are from different institutions, horizontal federated learning is adopted in this paper.

B. BiSeNet

Most current real-time semantic segmentation algorithms achieve faster speed by sacrificing accuracy, but the practical application effect is not good. BiSeNet, a two-sided segmentation network, can effectively balance the accuracy and speed of semantic segmentation. The network has broad receptive fields and rich spatial feature information by constructing two feature extraction paths to extract high-dimensional nonlinear features and low-dimensional spatial features. The architecture is mainly composed of the spatial path and context path.

The function of a spatial path is to maintain the size of the input image and obtain spatial information. The function of the context path is to obtain a sufficiently large receptive field to judge the target category, and it uses a series of attention modules (AM) to optimize the output contextual semantic feature information. The output features of the two paths are fused and pixel-level classified by the feature fusion module (FFM). Finally, the segmentation results are obtained by bilinear interpolation up-sampling. Regarding segmentation speed, although the image size of the spatial path input is large, it only has three layers of convolution block,

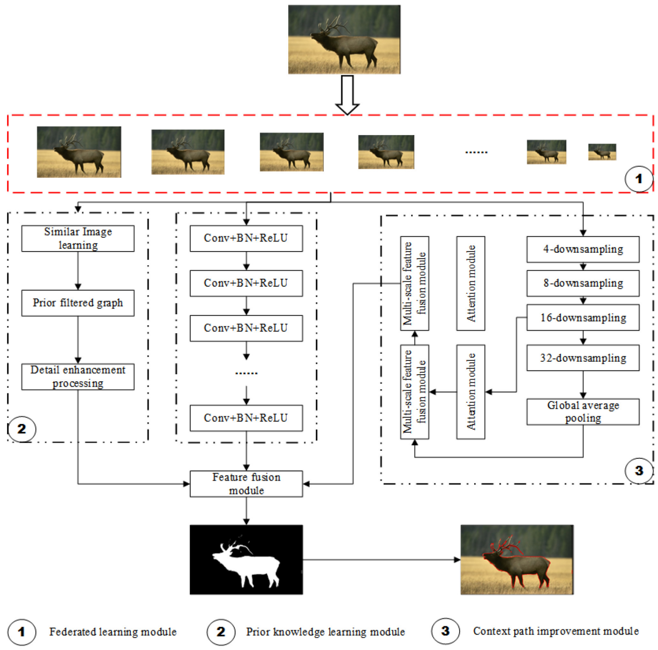


Fig. 2. Structure of proposed FLPK-BiSeNet for image edge extraction.

so the computational burden is small, while the context path adopts a lightweight network, which can quickly realize downsampling. In addition, the two modules can be calculated in parallel to improve the segmentation speed further. Regarding segmentation accuracy, spatial path output features have rich spatial detail information, while context path has a wide receptive field, which can fully extract image context information, so the network segmentation accuracy is high.

III. The Proposed FLPK-BiSeNet

Fig. 2 shows the overall structure of the network in this paper. The network consists of three main parts: the federated learning process, prior knowledge learning, spatial path and context path. The role of federated learning is to balance data and enrich image features. The role of prior knowledge learning is to improve image quality and optimize image edge extraction. The function of a spatial path is to maintain the size of the input image and obtain spatial information. The role of the context path is to obtain a sufficiently large receptive field and extract high-dimensional nonlinear features. The output of the network is obtained by the up-sampling module, which fuses the features of the two channels and up-sampling the feature graph obtained by convolution by 4-bilinear interpolation.

A. Federated Learning for Enriching Features

In each round of generation selection of federated learning, independent participants conduct local model training and upload the trained parameters to the base station. The central server completes parameter aggregation and update and sends the updated parameters to each participant to start a new round of iteration until the training converges.

Federated Average Algorithm (FedAvg) [31] is the most commonly used aggregation algorithm for federated learning. FedAvg can achieve cooperative training of local models by a certain number of global generation selections. For each global generation selection, assuming that the number of participants is *N*, the total number of samples is *D*, and the sample number of participants *k* is D_k, then the objective function to be optimized is $min f(w)(w \in R^d)$, where,

$$f(w) =^{def} \frac{1}{D}\sum_{j=1}^{D} f_j(w). \tag{1}$$

$$f_j(w) = l(u_j, v_j; w), \tag{2}$$

where $f_j(w)$ is the loss prediction of model parameter *w* on the $j - th$ sample data (u_j, v_j). For participant *k*, $F_k(w)$ is defined as:

$$F_k(w) = \frac{1}{D}\sum_{j \in p_k} f_j(w), \tag{3}$$

where p_k is the data distribution of participant *k*. Then the total loss function of the federated learning model can be expressed as:

$$f_w = \sum_{k=1}^{N} \frac{D_k}{D} F_k(w). \tag{4}$$

When the gradient of participant *k* is $g_k = \nabla F_k(w)$ and the learning rate is *R*, the updated parameter after *t* iterations is:

$$w_{t+1} = w_t - R\sum_{k=1}^{N} \frac{D_k}{D} g_k. \tag{5}$$

Then the local model parameter update mode of each participant is:

$$w_{t+1,k} = w_{t,k} - R\nabla F_k(w_k). \tag{6}$$

1) Client Training: We assume that the set of clients is $D = 1, 2, \ldots, N_D$. The $i - th$ client has one data set denoted as S_i, containing N_i samples. The node device set is $M = (1, 2, \ldots, N_M)$. Client D_i uploads the updated local model to its connected M_j. *S* represents the sum of all client devices, $S = \bigcup_{i=1}^{N_D} S_i$, and it contains N_S samples in total. The objective of FL is to minimize the following loss function:

$$f(w) = \frac{1}{N_S}\sum_{i=1}^{N_D}\sum_{S_k \in S_i} f_k(w), \tag{7}$$

where, $f_k(w)$ signifies the loss function of BiSeNet.

The client layout abides by the most basic FL setup. SGD will train each client model. For each epoch, the local model of device D_i is iterated N_i times. In the process of the $t - th$ iteration of the $l - th$ epoch, the weight $w_i^{t,l}$ is:

$$w_i^{t,l} = w_i^{t-1,l} - \frac{\beta}{N_i} \times B \tag{8a}$$

$$B = \left[\nabla f_k\left(w_i^{t-1,l}\right) - \nabla f_k\left(w^l\right)\right] + \nabla f\left(w^l\right), \tag{8b}$$

where $\beta > 0$ is the step size. This paper uses w_i^l to represent the local weight after the last generation selection in the

l epoch. For example, $w_i^l = w_i^{N_t,l}$. Then w^l represents the global weight update in the $l-th$ epoch:

$$\nabla f\left(w^l\right) = \frac{1}{N_S} \cdot \sum_{i=1}^{N_D} \sum_{S_k \in S_i} \nabla f_k\left(w^l\right). \tag{9}$$

$$w^l = w^{l-1} + \sum_{i=1}^{N_d} \frac{N_i}{N_S}\left(w_i^l - w^{l-1}\right). \tag{10}$$

Device D_i uploads its weight and gradient w_i^l, $\nabla f_k(w^l)_{S_k \in S_i}$ to the server. When a node generates a new block, the client eclipses the new block from the miner and updates the local model by calculating w^l based on the aggregation results in the new block.

2) Server-Side Parameter Aggregation: On the server side, each client device has a server device (also known as a miner) associated with it. All miners connected to the client device as nodes together constitute the blockchain. After receiving the parameter updates uploaded by each client device, miners verify the authenticity of the parameter updates by comparing the sample size N_i and the corresponding local computation time $T_{local,i}^l$. Validated update parameters are stored in the miner candidate block. In FL, block and miner cross-validation is designed to ensure the dependability of each miner's exchange in the local updating. Each block contains a header and a body: the body holds device local model updates of D_i, such as $w_i^l, \nabla f_k(w^l)_{S_k \in S_i}$, and the local computation time $T_{local,i}^l$; The header stores the pointer information to the previous block, λ and PoW output. Here, λ is the block generation rate. PoW denotes proof of work. Each block size is $l + \nu' \cdot N_D$. l is the head length. ν' is the size of the updated overall model. Each miner has a candidate block containing model updates from associated local devices or miners. The filling process of candidate blocks continues until the candidate blocks reach a scheduled block size or the waiting time T_{wait} is exceeded.

Considering the simplicity and robustness of PoW, this paper adopts PoW as the consistency algorithm. When the input of the miner is changed, it can randomly generate a hash value. If the value is less than one threshold, the entire process stops. Once the miner successfully finds a hash value that meets the criteria, its candidate block becomes a new block. The block generation rate λ is used to represent the difficulty of PoW. If the target hash value is lower, the λ is smaller. FL can also use other consistent algorithms (such as PoS or BFT) but requires more complex preparation.

The newly generated block by the above operation is diffused to all the other miners. At this point, all miners must stop the PoW operation after taking over the generated block. In the local ledger, the new block will be added. When the first generated block is within the spread delay, another miner M_2 successfully generates blocks. Some miners may improperly add the second generated block to the local ledger. This situation is referred to as bifurcation in this article. In FL, a new iteration is started over once a fork occurs.

In addition, the blockchain network provides data rewards and mining rewards to clients and miners. The data rewards of the client are received from the interrelated miners. The number is N_i. When the miner M_j produces a block, the mining payment is acquired by BN. The number of mining rewards is proportional to the total data sample size of all associated devices, $\sum_{i=1}^{N_{M_j}} N_i$, where N_{M_j} is the number of devices associated with miner M_j. It is worth noting that FL can further improve the reward mechanism by considering the size of the sample number and the quality of the data samples. **Algorithm 1** gives the single iteration flow of FL.

The server initializes model parameters, learning rate, iteration times, etc. Then the client trains a BiSeNet model, accepts the parameters from the server to initialize the global BiSeNet model, and starts training. The client sends the trained parameters to the server. The server uses the mediator to filter the parties, which can improve accuracy and update the weight of each party aggregated on the server. The aggregated model parameters are then assigned to the participants for the next round of training until the model converges, the model is saved, and the federated learning is finished. In addition, the mediator is a virtual component that can be deployed directly on the federated server to reduce communication overhead. The specific process is shown in **Algorithm 2**.

B. Prior Knowledge Learning

The similar image *R* is divided into several fragments and represented by $P_1, P_2, \ldots, P_k \in R$. Let noise modeling expression be $q = p + n$, p denotes the detail part, and n denotes the contaminated detail part. The learning process in this paper can be defined as $\hat{p} = A \times q$, where *A* represents the prior knowledge obtained by learning. To ensure the effectiveness of the eigendecomposition, let *A* be a symmetric matrix that satisfies the following:

$$A = U \Lambda U^T, \tag{11}$$

where, $U = (u_1, u_2, \ldots, u_k) \in R$ denotes the feature vector of each fragment. $\Lambda = diag(\lambda_1, \lambda_2, \ldots, \lambda_k) \in R$ is the eigenvalues of each fragment. Therefore, the learning process of this algorithm is equivalent to finding the eigenvector *U* and eigenvalue Λ of each fragment in similar images. It can be seen that *U* is composed of fragment eigenvectors and satisfies the orthogonality condition.

$$U^T U = I. \tag{12}$$

To ensure that fragments highly similar to the image to be processed are found, image fragments with high similarity should be screened using the orthogonality of *U*.

$$P = (P_1, P_2, \ldots, P_k). \tag{13}$$

In the screening process, considering that the direction of image sparsity will affect pixels, a diagonal weight matrix is added [18]. The spatial shape adaptability is determined by weight $W_1 \in R$, and the fragment similarity strength is determined by weight $W_2 \in R$, which satisfies the expression:

$$minU\left(||U^T W_1 P W_2||_\sigma\right), U^T U = I, \tag{14}$$

where σ represents the measure of group sparsity, aiming to screen out fragments similar to the image to be processed.

Algorithm 1 FL Single Iteration

```
1: for Local device parameters updating do
2:    Device D_i computes its own parameter w_i^{t,l} through N_i iterations;
3: end for
4: for Uploading local model parameters do
5:    Device D_i is randomly connected to miner M_i; Device D_i uploads its own parameter w_i^l, ∇f_k(w^l)_{S_k∈S_i} and local computation time T^l_{local,i} to the connected miner;
6:    IoU^i = IoU_t^i;
7: end for
8: for Cross-validation do
9:    Each miner broadcasts to other miners the received parameter updates from the local device. Validated local updating will be stored in the miner's candidate blocks until the candidate blocks reach the block size l+ν'·N_D or exceed the maximum waiting time T_{waitfor};
10: end for
11: for Block generation process do
12:    Each miner starts running the PoW until the target situation is found or a generating block is received from another miner's broadcast;
13: end for
14: for Block propagation do
15:    M_x ∈ M represents the first miner to reach the target. The candidate block of M_x is treated as a new block and broadcast to all miners. To avoid bifurcation, an ACK frame is set to indicate whether bifurcation occurs (1 indicates bifurcation; 0 indicates no bifurcation); they will be sent when each miner receives a new block. If a bifurcation occurs, it starts over from step 1. A miner generating a new block waits until a predefined maximum waiting time T_{a,wait} for ACK frames is reached;
16: end for
17: for Global model download do
18:    Local device D_i obtains a neo-generated block from the connected miner;
19: end for
20: for Global model update do
21:    Local device D_i computes global parameter update based on the aggregated results;
22: end for
23: The above procedure runs iteratively until w^l - w^{l-1} ≤ ε. As seen from step (7), global model updates are computed locally on each device in FL. This also ensures that miner or server failures do not affect local updates to the global model and prevents excessive computational overhead for miners.
```

And to calculate it simply, let $W_1 = I$ and define:

$$W_2 = \frac{1}{\eta} diag\left(e^{-||q-p_1||^{2/h^2}}, \ldots, e^{-||q-p_k||^{2/h^2}}\right), \quad (15)$$

where η represents the normalization constant, and h represents the sparsity decay parameter. The feature decomposition

Algorithm 2 Federated Learning Based on BiSeNet Model

```
Input: global weight, local generation times, local data, learning rate;
Executing on the server
Initialize the original model parameter W_0 and broadcast the original model parameter W_0 to all participants.
for Each global model update round t = 1, 2, ... do
   The regulator determines the set of C participants by screening.
end for
for All participants in parallel do
   Update model parameters locally: w_{t+1}^k ← participants update;
   The updated model parameter w_{t+1}^k is sent to coordinate;
end for
The received model parameters are aggregated; that is, a weighted average is used for the received model parameters: G^{t+1} = G^t + φ Σ_{i=1}^m (L_i^{t+1} - G_i^t).
for Model convergence do
   Send a signal to the coordinator so that all w_{t+1}^{Avg} stop training and will send the model parameters to it.
end for
Executing on the regulator
Get the participants' latest local model parameters and test sets on the server.
for All participants C do
   if D[f(x; w_{t+1}), y] > D[f(x; w^k), y] - d
   Store the available model parameters into C_t;
   End if
   Determine the parameters and quantity C_t of stored participating and send them to the server.
end for
Update on participants
The participant trains the model locally and maintains q_τ(x, w)(c) = exp(z_c/τ) / Σ_{i=1}^c exp(z_c/τ).
Get the latest model parameters from the server, i.e., let w_t^k = w_t^{Avg}.
for Each local iteration from one to S do
   Computing the gradient: w_{t+1}^{Avg} = Σ_{k=1}^K (n_k/n) w_t^k - (η/n) ∇_{w_t^k} Σ_{i=1}^{n_k} f((x^i, w_t^k), y^i);
   Update model parameters locally: w_{t+1}^k = w_t^k;
end for
Get the local model parameter update w_{t+1}^k and send it to the coordinator.
```

results of the final screening fragments are satisfied:

$$[U, S] = eigen\left(PP^T\right), \quad (16)$$

where U is eigenvector matrix. S is the eigenvalue matrix.

The estimated eigenvalue Λ is composed of the eigenvector value λ_1 of each similar fragment, and its expression is:

$$\lambda_i = \frac{\left(u_i^T p\right)^2}{\left(u_i^T p\right)^2 + \sigma^2}, i = 1, 2, \ldots, k, \quad (17)$$

where p represents the initial estimate of image details. Due to the uncertainty of the initial estimate, this paper uses the minimized Bayesian mean squared error (BMSE) [32] to judge the different detail fragments, and the expression satisfies:

$$BMSE = E_p\left[E_{q|p}\left[||U\Lambda U^T q - p||_2^2 | p\right]\right], \tag{18}$$

where conditional distribution $f(q|p)$ satisfies Gaussian independent and identical distribution. We can obtain the following:

$$f(q|p) = N\left(p, \sigma^2, I\right). \tag{19}$$

The prior distribution $f(p)$ is defined according to the uniqueness of k matching fragments $p_1, p_2, \ldots, p_k$, and then the mean μ and covariance Σ of $f(p)$ are obtained, its expression meets:

$$\mu = \sum_{i=1}^{k} \varpi_i p_i. \tag{20}$$

$$\Sigma = \sum_{i=1}^{k} \varpi_i (p_i - \mu)(p_i - \mu)^T, \tag{21}$$

where the weight ϖ_i expression is:

$$\varpi_i = \frac{1}{\eta} e^{-||q-p_i||^{2/h^2}}. \tag{22}$$

Substituting Equation (22) into Equations (20) and (21), the minimum Bayes of eigenvalue Λ can be obtained, and the expression is:

$$\Lambda = argminBMSE = \frac{diag\left(U^T \Sigma U + U^T \mu \mu^T U\right)}{diag\left(U^T \Sigma U + U^T \mu \mu^T U\right) + \sigma^2 I}. \tag{23}$$

The final expression of Λ is:

$$\Lambda = \frac{S}{S + \sigma^2 I}. \tag{24}$$

Next, we enhance the detailed features of the image. Let image $\hat{p}$ be a global variable $F(x)$, where x represents the gray value in the image. Let the image $\hat{p}'$ after local uniform sparse processing be a global variable $F(y)$, and y represents the gray value in the image after local uniform sparse processing. The enhancement process satisfies $y = T(x)$. The solution process of $T(x)$ is as follows:

$$F(y) = F(x)|_{x=T^{-1}(y)}, \tag{25}$$

where $T^{-1}(y)$ is the inverse of $T(x)$. Taking the derivative of both sides of this equation, we can get:

$$f(y) = f\left[T^{-1}(y)\right] \cdot \frac{d\left[T^{-1}(y)\right]}{dy} \tag{26a}$$

$$= f(x) \cdot \frac{dx}{dy}, \tag{26b}$$

where $f(x) = \frac{n_k}{MN}$, n_k represents the proportion of pixels in all pixels. M and N represent the rows and columns of the image, respectively. $f(y) = \frac{1}{L-1}$, L represents the number of gray levels.

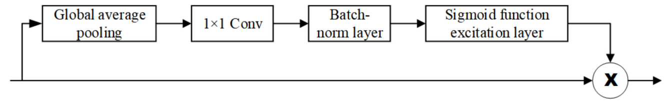


Fig. 3. AM structure.

Substituting the results of $f(x)$ and $f(y)$ into the above equation, and we get:

$$\frac{1}{L-1} = \frac{n_k}{MN} \cdot \frac{d(x)}{d[T(x)]}. \tag{27}$$

$$d[T(x)] = \frac{(L-1) \cdot n_k}{MN} dx. \tag{28}$$

By integrating both sides of the formula, the relation of enhanced texture can be obtained as:

$$T(x) = \frac{L-1}{MN} \sum_{j=0}^{k} n_{j'}. \tag{29}$$

Image detail feature enhancement expression satisfies:

$$p_z = \hat{p}' - \hat{p}, \tag{30}$$

where p_z represents the image with enhanced detail features.

C. Novel Context Path Structure

The main function of a context path is to provide sufficient receptive field and semantic information. In the BiSeNet model's context path, the lightweight model Xception can down-sample the feature map quickly to obtain large receptive fields. The receptive field encodes high-level semantic context information. An average global pool is then added to the tail of the lightweight model, which can provide global context information for the maximum receptive field. Finally, the network combines the up-sampled output characteristics of the global pool with the lightweight model. The context path structure is shown in Fig. 2.

In the context path, BiSeNet also designs the attention module (AM) to optimize the features in each stage. As shown in Fig. 3, AM uses global average pooling to capture global context and computes attention vectors to guide feature learning. The AM optimizes the output features of each stage in the context path. Global context information can also be easily integrated without any up-sampling operations. Therefore, the computational cost is negligible.

Because the output of the spatial path is a low-level spatial information feature, while the output of the context path is a high-level contextual information feature, BiSeNet designs a special feature fusion module to fuse these features. Given the different levels of features, BiSeNet first connects the spatial and context paths' output features. Then batch normalization is used to balance the scale of features. Next, the connected features are pooled into feature vectors and the weight vectors are calculated. This weight vector can reweight features, equivalent to feature selection and combination functions. Fig. 4 shows the structure of this design.

In the original context path structure, the low-level and deeper high-level abstract semantic feature information of the

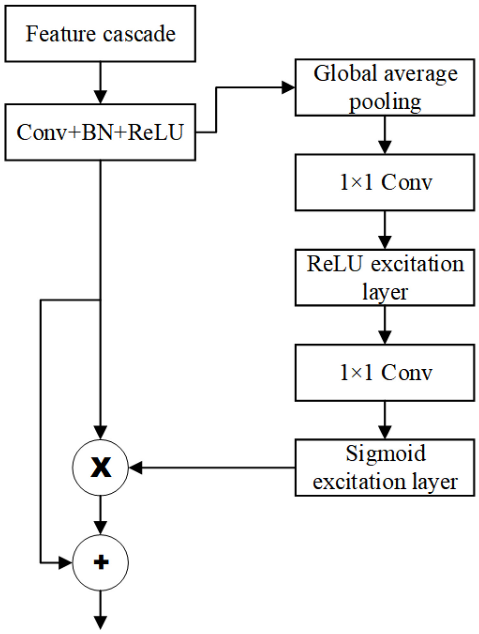


Fig. 4. Special feature fusion module.

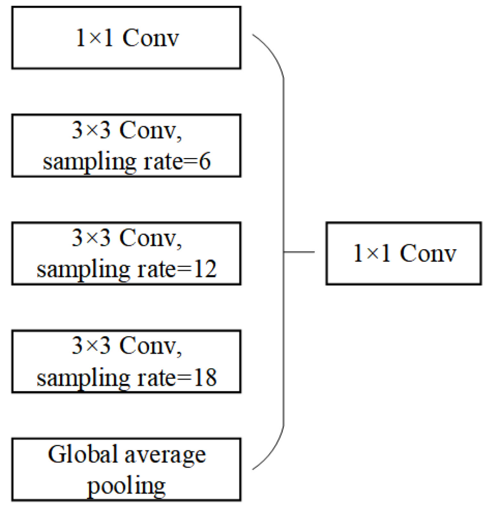


Fig. 5. Model of void pyramid pooling layer.

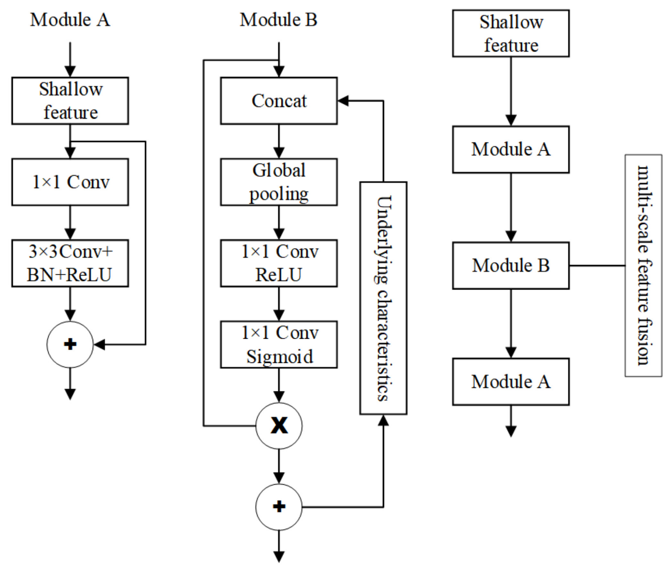


Fig. 6. Structure of multi-scale feature fusion.

Xception model is used for concatenation. This deep abstract feature has a strong identification of the object's size, Angle and position information, and the invariance of translation position, rotation Angle and scale size plays a key role in improving the segmentation performance of each object in the image.

However, because the deep abstract features have a large receptive field, too strong semantic information will implicitly lose the spatial information of some objects in the image. As a result, the advantage of positioning accuracy of the deep network needs to be recovered. However, if the deep abstract features are directly up-sampled, the edge details of object segmentation will suffer a great loss. Therefore, integrating multi-level and multi-scale feature information can complement the shallow local details and deep abstract features and get a better segmentation effect. However, if channel connection is carried out directly, the differences in features in different stages should be addressed, leading to consistent segmentation targets. Therefore, the context path is improved. Firstly, a dilated pyramid pooling layer [6] is added to the end of the Xception model to obtain the underlying abstract features. Then, the multi-scale feature fusion module designed in this study gradually fuses from the deep layer to the shallow layer. The final fused and spatial channel features are input into the feature fusion module for final segmentation.

In the modified context path, a dilated pyramid pooling is used at the end of the Xception model to process the underlying features. It includes one convolution operation with a convolution kernel size of 1 × 1, three convolution operations with a convolution kernel size of 3 × 3, three dilated convolution operations with sampling rates of 6, 12, and 18, and one global average pooling operation. Then, the feature layers obtained by each operation are connected through channels. After the convolutional layer with a convolution kernel of 1 × 1 size is processed, it is the output feature of dilated pyramid pooling. The model structure is shown in Fig. 5.

The structure of the multi-scale feature fusion module is shown in Fig. 6. The multi-scale feature fusion module has two small modules, A and B. The main function of module A is to refine feature information, and the main function of module B is to carry out feature fusion. The shallow information first goes through 1 × 1 convolution in module A to reduce the number of channels, which is convenient to improve the computational ability of the model. Then the residual module combines the channel information to refine the feature. The refined shallow features and deep features are input into module B. After channel addition, a global average pooling operation is carried out, and two 1 × 1 convolution operations and incentive operations are carried out. The obtained feature map is multiplied with shallow features and then added with deep features. Semantic selection of shallow features is carried out through deep feature information to obtain features with more discriminative power. Then the output of module B is input into module A again for feature refinement; that is, A feature fusion operation is completed.

D. Loss Function

In image edge detection, the problem of edge point loss classification is usually considered as the problem of dichotomous loss cost between edge points and non-edge points [33]. Therefore, the cross entropy classification problem of label image is taken as the loss cost function. To improve the robustness of the edge detection model, the threshold method is utilized to normalize the pixel values in the label graph, and the label graph is transformed into the probability graph of edge information. The pixels whose probability value is greater than the threshold are regarded as edge points and the controversial pixels are excluded. The loss function is expressed as:

$$l(X_i; W) = \begin{cases} \varphi \times log[1 - P(X_i; W)], & y_i = 0, \\ 0, & 0 < y_i \leq \eta, \\ \phi \times logP(X_i; W), & y_i > \eta, \end{cases} \quad (31)$$

where,

$$\varphi = \lambda \cdot \frac{|S^+|}{|S^+| + |S^-|}. \quad (32)$$

$$\phi = \lambda \cdot \frac{|S^-|}{|S^+| + |S^-|}, \quad (33)$$

where X_i represents the activation value of the neural network. W represents learnable parameters in the neural network. y_i represents the probability that the label graph pixel is the edge point. The hyperparameter λ balances the difference in the number of positive and negative samples. $|S^+|$ and $|S^-|$ are the number of positive and negative samples, respectively.

The edge images output by each stage in the network are quite different from each other, and the magnitude of loss in each stage may be inconsistent. The loss in the fusion stage should be the main position. To balance the relationship between the loss of each stage and the fusion loss, the loss proportion of the five stages in the network is reduced, and the loss proportion of the fusion stage is increased. The total loss function can be expressed as:

$$L(W) = \sum_{i=1}^{|I|} \left[\sum_{k=1}^{K} S^k l\left(X_i^k; W\right) + l_{fuse} l\left(X_i^{fuse}; W\right) \right], \quad (34)$$

where the loss weight of the fusion layer is l_{fuse}. S^k represents the loss weight of the $k-th$ stage. X_i^k represents the excitation value of the $i-th$ pixel in the output graph of the $k-th$ stage. X_i^{fuse} represents the excitation value of the $i-th$ pixel in the fusion module. $|I|$ represents the total pixel number in each image. K represents the number of stages in the backbone network.

IV. Experiment and Result Analysis

In the actual training of the network, the spatial paths use three convolutional layers with convolution kernels of size 7×7, 3×3 and 3×3, respectively. The context path uses the Xception39 network model. The optimization algorithm is stochastic gradient descent (SGD) [34]. This optimization algorithm is simple, efficient and requires less memory. Batch size=1, the momentum is 0.9, and the weight decay is $1e^{-4}$. The learning rate decay strategy is also used to control the learning rate, where the initial learning rate η is $1.8e^{-2}$. The updating formula for each generation of learning is as follows:

$$\eta = \eta \times \left(1 - \frac{iter}{max - iter}\right)^{mo}, \quad (35)$$

where *iter* is the number of current iterations, *max-iter* is the total number of iterations, which is set as 250 times (i.e., 250 epochs) in this study. *mo* is set to 0.8; The balance parameter α of the primary and secondary loss is set to 1. Here we make a brief training caparison with different methods. The result is shown in Table I.

TABLE I
RESULTS OF TRAINING METHODS

Model	Error rate/%
AdaGrad	12.7
RMSProp	9.6
Backpropagation	3.2

The experiment is mainly implemented based on MATLAB programming language, the computer operating system is Win10, the environment configuration is CUDA9.2, torch1.3, and the computer graphics card is NVIDIA GTX Geforce 1060Ti. The experiment is carried out on nature images. Accuracy and running time are used as evaluation indexes.

A. Ablation Experiments

The proposed FLPK-BiSeNet method contains three parts: Federated Learning, Priori Knowledge and BiSeNet. We conducted an ablation experiment to verify each part's effect on the performance results of the overall image edge extraction. The results are shown in Table II and Fig. 7.

TABLE II
RESULTS OF ABLATION EXPERIMENT

Model	Accuracy/%	Time/s
BiSeNet(A)	82.5	1.2
BiSeNet+Federated Learning(B)	92.5	1.4
BiSeNet+Priori Knowledge(C)	93.7	1.3
Priori Knowledge+BiSeNet+Federated Learning(D)	98.4	1.5

Table II shows that the single BiSeNet only achieves 82.5% accuracy. The accuracy values of BiSeNet+Federated Learning and BiSeNet+Priori Knowledge are 92.5% and 93.7%. For proposed FLPK-BiSeNet achieves 98.4%, which improves by 4.7%, 5.9% and 15.9%. In terms of running time, although the time of FLPK-BiSeNet is 1.5s, the time of the three models is almost the same. Comprehensive analysis shows that the FLPK-BiSeNet model has better advantages.

B. Performance Comparison of Loss Function

To verify the availability of the FLPK-BiSeNet loss function, the cross-entropy loss function commonly used in image segmentation and the proposed loss function are selected for comparative experiments. The segmentation results of BiSeNet

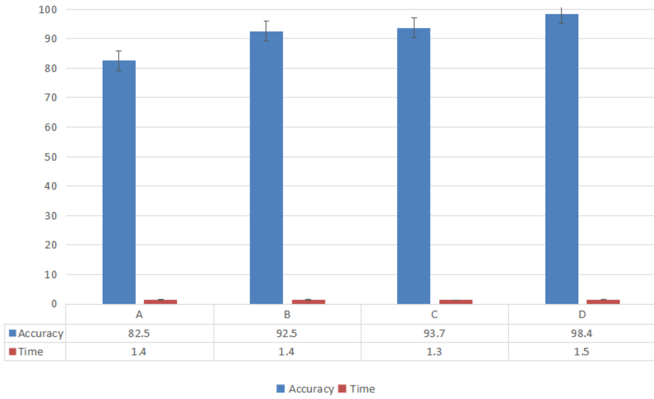
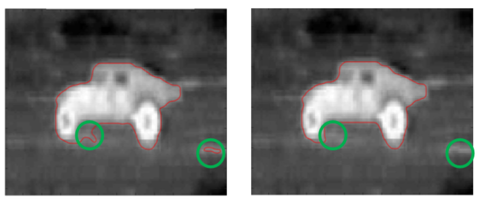


Fig. 7. Visualization results.



(a) BiSeNet with cross-entropy (b) Proposed

Fig. 8. Comparison of segmentation results by different loss functions.

TABLE III
ACCURACY COMPARISON WITH DIFFERENT LOSS FUNCTIONS

Loss function	Cross entropy	Proposed
Accuracy/%	91.7	94.8

with cross-entropy loss function and the proposed network are shown in Fig. 8(a) and Fig. 8(b).

According to the extraction results, the extraction results of the network using cross-entropy loss in the front wheel region are not ideal, and there are some holes in the background region of the BiSeNet result. The network can learn more complex scenes using the edge enhancement loss function, such as the junction area between the front wheel and the background. This method can better extract the contour of the front wheel boundary.

Table III counts the evaluation indexes of the edge extraction results of the above experiments on the test dataset. As seen from Table III, the edge enhancement loss function improves the proposed method's segmentation performance and accuracy by 3.1%. The above results prove that the proposed edge enhancement loss function can increase the learning proportion of the network to the target region and effectively enhance the ability of the network to extract the edge region.

C. Comparison With Other Methods

To fully verify the edge extraction performance of FLPK-BiSeNet, the traditional C-V model, LIF, BiSeNet and the latest methods EPS [35], GA [36], DCP [37], PICA [38], DLE [39] are selected for comparison. The test data set aircraft and cells are selected as input images in the experiment, and the results are shown in Figs. 9, 10, and 11.

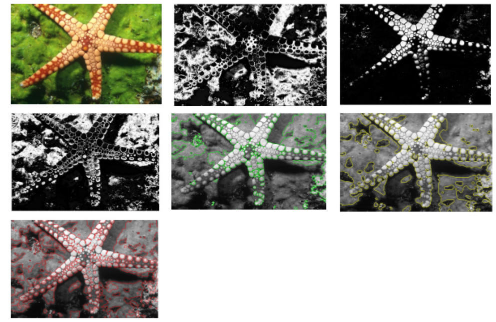


Fig. 9. Results with different methods. From left to right. Input image, C-V, BiSeNet, DCP, PICA, DLE and the proposed model.

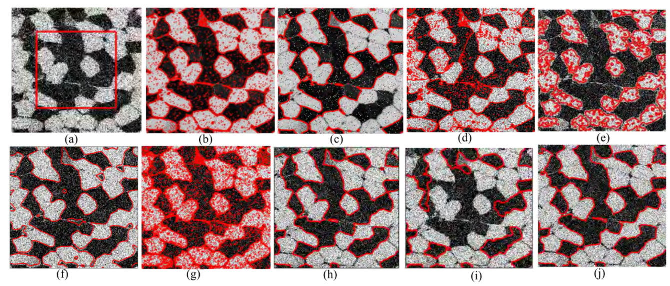


Fig. 10. Results with different methods. Row 1: Input image with Gaussian noise=0.05; (a)-(j): C-V, LIF, BiSeNet, EPS, GA, DCP, PICA, DLE, and the proposed model.

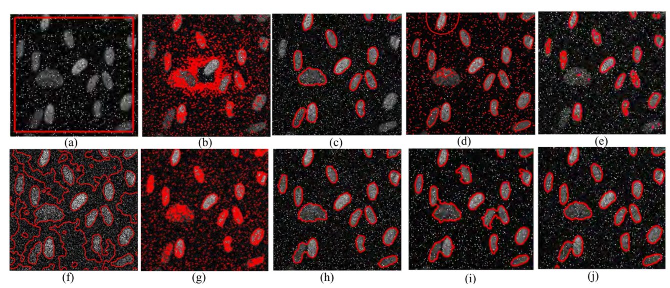


Fig. 11. Results with different methods. Row 1: Input image with Gaussian noise=0.05; (a)-(j): C-V, LIF, BiSeNet, EPS, GA, DCP, PICA, DLE, and the proposed model.

When the traditional method extracts the edge, the edge extraction needs to be completed. The C-V model has the worst extraction effect. It can be seen from the above figures that there are some blurred lines in the edge image generated by the traditional method, and there is a large amount of background noise in the feature map. The proposed method can extract the edge of the image and deal with the fuzzy problem of some details. It can effectively pay attention to the key edge information of the image and reduce the influence of irrelevant information on the image edge extraction. In practice, high-quality similar images can be easily obtained from cell images. Therefore, this paper increases the degree of noise pollution for such images, further highlighting the proposed algorithm's anti-noise performance. Fig. 10 is a cell image polluted by Gaussian noise with a standard deviation of 0.05. Compared with Fig. 10(b,c), the traditional edge detection algorithm has a poor effect. They extract cell edge information and noise edge together, which seriously interferes with the recognition

TABLE IV
COMPARISON RESULTS OF FIGURE 9

Model	Accuracy/%	Time/s
C-V	68.9	5.4
LIF	70.2	4.9
BiSeNet	79.5	1.1
EPS	89.7	2.8
GA	92.3	2.4
DCP	93.6	2.6
PICA	94.8	2.2
DLE	95.2	1.6
FLPK-BiSeNet	98.8	1.2

TABLE V
COMPARISON RESULTS OF FIGURE 10

Model	Accuracy/%	Time/s
C-V	66.5	4.8
LIF	69.7	4.6
BiSeNet	77.4	1.2
EPS	88.6	3.1
GA	92.3	2.3
DCP	94.1	2.4
PICA	95.5	2.7
DLE	96.6	2.0
FLPK-BiSeNet	98.1	1.4

TABLE VI
COMPARISON RESULTS OF FIG. 11

Model	Accuracy/%	Time/s
C-V	64.2	5.7
LIF	63.5	5.1
BiSeNet	74.9	1.1
EPS	79.3	4.3
GA	88.4	3.4
DCP	92.9	3.0
PICA	94.7	2.5
DLE	96.1	1.8
FLPK-BiSeNet	98.3	1.3

of image edge information. However, the proposed algorithm can detect the cell target edge. However, the defect is that some edge textures still appear wobbly.

Compared with the proposed method, other advanced methods contain background noise of non-key edge information in each stage, and the edge lines are blurred. The proposed method utilizes the ability of federated learning to make image edges clearer by prior knowledge learning and convolutional network cross-layer feature fusion at different levels. It enables the overall network to pay attention to the edge information of key regions in the global environment, which helps to integrate multi-scale features fully. Compared with other advanced methods, the edge map in the output stage of the proposed method also reduces the input with some irrelevant edge information, especially in stage 4 and stage 5, without too much background clutter texture, which further verifies that the proposed method improves the extraction effect of image edge information.

Tables III, IV, and V count the evaluation indexes of the extraction results with different methods on different test datasets under the same experimental environment. Meanwhile, Figs. 12, 13, and 14 exhibit the visual results.

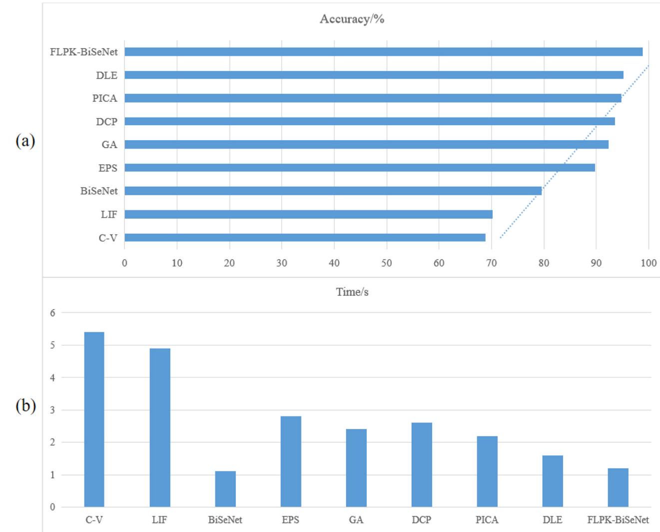


Fig. 12. Visual result of Table III: (a) accuracy, (b) time.

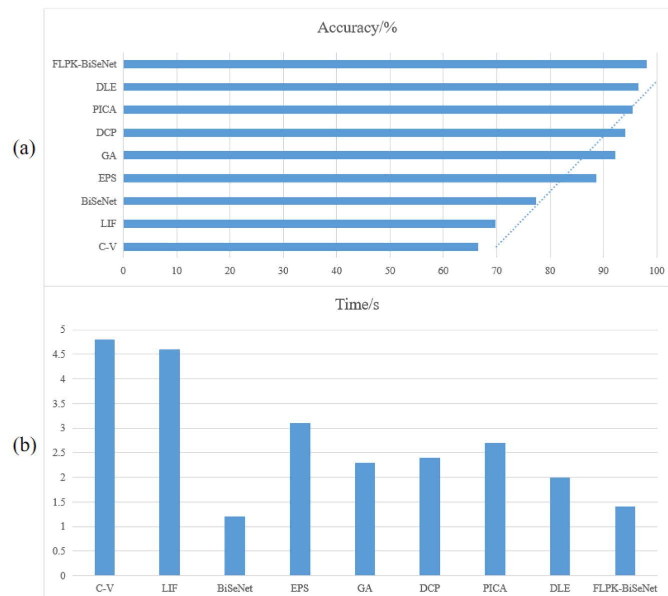


Fig. 13. Visual result of Table IV: (a) accuracy, (b) time.

V. CONCLUSION

Aiming at the problems existing in image edge extraction, this paper proposes a federated learning approach based on Priori knowledge and the bilateral segmentation network. It consists of three parts: federated learning, prior knowledge learning and context path learning. A new improvement is made in each module to reduce the processing area of the algorithm further, improve the overall detection speed, and strengthen the ability of network feature extraction. Experiments are carried out based on natural image data, the performance of the FLPK-BiSeNet method is compared with the current mainstream edge extraction network framework, and the effectiveness of the proposed method is verified. Experimental results show that the proposed method has better extraction accuracy and faster processing speed. In future

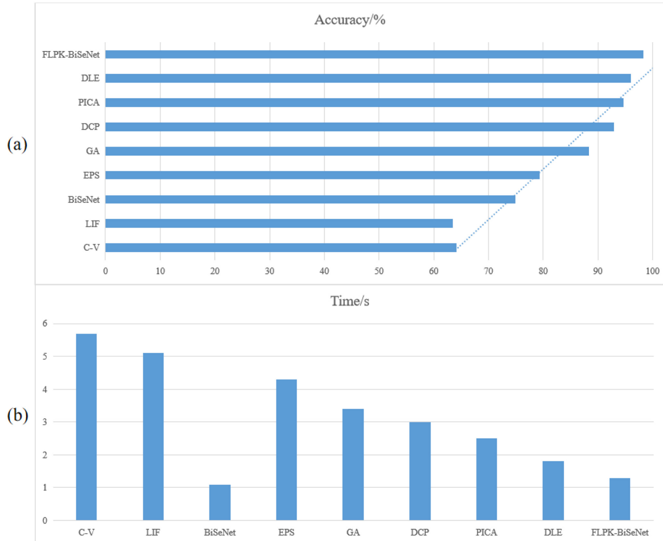


Fig. 14. Visual result of Table V: (a) accuracy, (b) time.

works, we will research more advanced deep learning methods and apply them in aerospace fields.

Conflicts of Interest

The authors declare that they have no conflict of interest with respect to the research, authorship and/or publication of this article.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Authors Contributions

All the authors made contributions to the article in different areas. Conceptualization, L.T., Y.Q. and M.S.; investigation, Y.Q. and G.S.; simulation, A.R.J., T.R.G. and S.Y.; writing and original draft preparation, L.T. and M.S.; writing, review and editing, Y.Q., and M.S. All authors have read and agreed to the published version of the manuscript.

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